

Section Problems Week 4

1) How do we calculate the following centrality measures?

A closeness centrality measure is calculated by:

Closeness centrality is the nearness to others. Closeness centrality is calculated by....adding up the shortest paths between two nodes, take the reciprocal, then multiply by the number of nodes N . *How far on average is a node from any other node.*

A betweenness centrality measure is calculated by:

Betweenness centrality is how many shortest paths between two nodes i and j is node k a part of. It is measuring brokerage.

An out-degree centrality measure is calculated by:

Out degree centrality is measured by adding up all the edges that a node sends out. Used in directed graphs/asymmetric matrices

A degree centrality measure is calculated by:

Measures the popularity. Degree centrality measures how many other nodes a particular node is connected to. Undirected graphs/symmetric matrices.

An in-degree centrality measure is calculated by:

The converse of out-degree centrality. How many nodes does a particular node receive nominations from. Used in directed graphs/asymmetric matrices.

An eigenvector centrality matrix is calculated by:

MATH (Eigenvector decomposition)- so let's not worry about the particulars. A node derives centrality from the centrality of its adjacent nodes. A nodes centrality increases with the centrality of its adjacent nodes. Having important connections makes a node important, not necessarily having lots of connections.

2) What do each of the concepts of degree centrality, betweenness centrality, closeness centrality, and eigenvector centrality seek to measure?

Degree centrality- Popularity

Betweenness centrality- Brokerage

Closeness centrality- Nearness to others

Eigenvector centrality- Status

3) What is the closeness centrality of the nodes in the matrix?

There is an error in this question. I provided an adjacency matrix. Closeness centrality is calculated with a distance matrix. However, the logic of calculation is still the same, so the answer would be found in the same logical way.

	A	B	C	D	E	F
A	--	1	0	0	1	0
B	1	--	1	0	1	1
C	0	1	--	0	0	1
D	0	0	0	--	0	0
E	1	1	0	0	--	1
F	0	1	1	0	1	--

A-2 .5 *6 3

B-4 .25 *6 1.5

C-2 .5 *6 3

D-0 Undefined *6 Undefined There is no such thing as a 0 in a reachability matrix though so this is no actually going to happen, you'll get infinity as below.

E3 .33 *6 2

F-3 .33 *6 2

Here's an actual reachability matrix and calculation of closeness centrality

	A	B	C	D	E	F
A	0	1	1	2	1	0
B	1	0	1	1	2	0
C	1	1	0	1	2	0
D	2	1	1	0	3	0
E	1	2	2	3	0	0
F	Inf	Inf	Inf	Inf	Inf	Inf

Step 1) Add across the columns for each row A

Step 2) Take the reciprocal of each of the summed values

Step 3) Multiply by the number of nodes, the result is a nodes closeness centrality score

A- 5 .2 *6 1.2

B- 5 .2 *6 1.2

C- 5 .2 *6 1.2

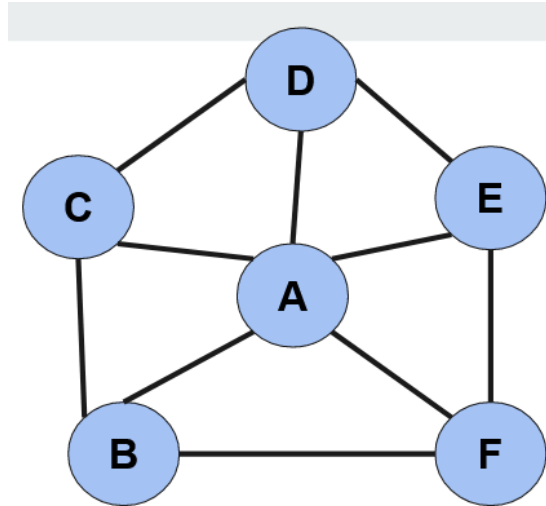
D- 7 .14 *6 .84

E- 8 .125 *6 .75

F- Inf 0 *6 0

Remember 1/Inf is 0.

4) What is the clustering coefficient of the following graph, with ego as Node A?



$$\frac{m}{\frac{n(n-1)}{2}} = \frac{5}{\frac{5(5-1)}{2}} = \frac{5}{\frac{20}{2}} = \frac{5}{10} = .5$$

To calculate the clustering coefficient, which is the density of the ego network, you remove the ego and all ties that ego shares with alters. Thus, we remove node A, and edge AC, AD, AE, AB, and AF. We then calculate the density of the ties that connect alters. In this case, there are 5 edges and 5 nodes. We use the same density equation as for an undirected graph in this case as the ego-network is composed of symmetric ties in this case.

